17.6.1 A Generic Example 347
17.6.2 Six Sigma for Software Engineering 349
17.7 Software Reliability 350
17.7.1 Measures of Reliability and Availability 350
17.7.2 Use of AI to Model Reliability 351
17.7.3 Software Safety 352
17.8 The ISO 9000 Quality Standards 353
17.9 The SQA Plan 354
17.10 Summary 355

CHAPTER 18 SOFTWARE SECURITY ENGINEERING 356

18.1 Why Software Security Information Is Important 357
18.2 Security Life-Cycle Models 357
18.3 Secure Development Life-Cycle Activities 359
18.4 Security Requirements Engineering 360
18.4.1 SQUARE 360
18.4.2 The SQUARE Process 360
18.5 Misuse and Abuse Cases and Attack Patterns 363
18.6 Security Risk Analysis 364
18.7 Threat Modeling, Prioritization, and Mitigation 365
18.8 Attack Surface 366
18.9 Secure Coding 367
18.10 Measurement 368
18.11 Security Process Improvement and Maturity Models 370
18.12 Summary 370

CHAPTER 19 SOFTWARE TESTING—COMPONENT LEVEL 372

19.1 A Strategic Approach to Software Testing 373
19.1.1 Verification and Validation 373
19.1.2 Organizing for Software Testing 374
19.1.3 The Big Picture 375
19.1.4 Criteria for “Done” 377

19.2 Planning and Recordkeeping 378
19.2.1 Role of Scaffolding 379
19.2.2 Cost-Effective Testing 379
19.3 Test-Case Design 381
19.3.1 Requirements and Use Cases 382
19.3.2 Traceability 383
19.4 White-Box Testing 383
19.4.1 Basis Path Testing 384
19.4.2 Control Structure Testing 386
19.5 Black-Box Testing 388
19.5.1 Interface Testing 388
19.5.2 Equivalence Partitioning 389
19.5.3 Boundary Value Analysis 389
19.6 Object-Oriented Testing 390
19.6.1 Class Testing 390
19.6.2 Behavioral Testing 392
19.7 Summary 393

CHAPTER 20 SOFTWARE TESTING—INTEGRATION LEVEL 395

20.1 Software Testing Fundamentals 396
20.1.1 Black-Box Testing 397
20.1.2 White-Box Testing 397
20.2 Integration Testing 398
20.2.1 Top-Down Integration 398
20.2.2 Bottom-Up Integration 399
20.2.3 Continuous Integration 400
20.2.4 Integration Test Work Products 402
20.3 Artificial Intelligence and Regression Testing 402
20.4 Integration Testing in the OO Context 404
20.4.1 Fault-Based Test-Case Design 405
20.4.2 Scenario-Based Test-Case Design 406
20.5 Validation Testing 407
20.6 Testing Patterns 409
20.7 Summary 409

CHAPTER 21 SOFTWARE TESTING—SPECIALIZED TESTING FOR MOBILITY 412

21.1 Mobile Testing Guidelines 413
21.2 The Testing Strategies 414
21.3 User Experience Testing Issues 415
21.3.1 Gesture Testing 415
21.3.2 Virtual Keyboard Input 416
21.3.3 Voice Input and Recognition 416
21.3.4 Alerts and Extraordinary Conditions 417
21.4 Web Application Testing 418
21.5 Web Testing Strategies 418
21.5.1 Content Testing 420
21.5.2 Interface Testing 421
21.5.3 Navigation Testing 421
21.6 Internationalization 423
21.7 Security Testing 423
21.8 Performance Testing 424
21.9 Real-Time Testing 426
21.10 Testing AI Systems 428
21.10.1 Static and Dynamic Testing 429
21.10.2 Model-Based Testing 429
21.11 Testing Virtual Environments 430
21.11.1 Usability Testing 430
21.11.2 Accessibility Testing 433
21.11.3 Playability Testing 433
21.12 Testing Documentation and Help Facilities 434
21.13 Summary 435

CHAPTER 22 SOFTWARE CONFIGURATION MANAGEMENT 437

22.1 Software Configuration Management 438
22.1.1 An SCM Scenario 439
22.1.2 Elements of a Configuration Management System 440
22.1.3 Baselines 441

22.1.4 Software Configuration Items 441
22.1.5 Management of Dependencies and Changes 442
22.2 The SCM Repository 443
22.2.1 General Features and Content 444
22.2.2 SCM Features 444
22.3 Version Control Systems 445
22.4 Continuous Integration 446
22.5 The Change Management Process 447
22.5.1 Change Control 448
22.5.2 Impact Management 451
22.5.3 Configuration Audit 452
22.5.4 Status Reporting 452
22.6 Mobility and Agile Change Management 453
22.6.1 e-Change Control 453
22.6.2 Content Management 455
22.6.3 Integration and Publishing 455
22.6.4 Version Control 457
22.6.5 Auditing and Reporting 458
22.7 Summary 458

CHAPTER 23 SOFTWARE METRICS AND ANALYTICS 460

23.1 Software Measurement 461
23.1.1 Measures, Metrics, and Indicators 461
23.1.2 Attributes of Effective Software Metrics 462
23.2 Software Analytics 462
23.3 Product Metrics 463
23.3.1 Metrics for the Requirements Model 464
23.3.2 Design Metrics for Conventional Software 466
23.3.3 Design Metrics for Object-Oriented Software 468
23.3.4 User Interface Design Metrics 471
23.3.5 Metrics for Source Code 473
23.4 Metrics for Testing 474
23.5 Metrics for Maintenance 476
23.6 Process and Project Metrics 476

23.7 Software Measurement 479
23.8 Metrics for Software Quality 482
23.9 Establishing Software Metrics Programs 485
23.10 Summary 487

PART FOUR MANAGING SOFTWARE PROJECTS 489

CHAPTER 24 PROJECT MANAGEMENT CONCEPTS 490

24.1 The Management Spectrum 491
24.1.1 The People 491
24.1.2 The Product 491
24.1.3 The Process 492
24.1.4 The Project 492
24.2 People 493
24.2.1 The Stakeholders 493
24.2.2 Team Leaders 493
24.2.3 The Software Team 494
24.2.4 Coordination and Communications Issues 496
24.3 Product 497
24.3.1 Software Scope 497
24.3.2 Problem Decomposition 497
24.4 Process 498
24.4.1 Melding the Product and the Process 498
24.4.2 Process Decomposition 498
24.5 Project 500
24.6 The W^5HH Principle 501
24.7 Critical Practices 502
24.8 Summary 502

CHAPTER 25 CREATING A VIABLE SOFTWARE PLAN 504

25.1 Comments on Estimation 505
25.2 The Project Planning Process 506

25.3 Software Scope and Feasibility 507
25.4 Resources 507
25.4.1 Human Resources 508
25.4.2 Reusable Software Resources 509
25.4.3 Environmental Resources 509
25.5 Data Analytics and Software Project Estimation 509
25.6 Decomposition and Estimation Techniques 511
25.6.1 Software Sizing 511
25.6.2 Problem-Based Estimation 512
25.6.3 An Example of LOC-Based Estimation 512
25.6.4 An Example of FP-Based Estimation 514
25.6.5 An Example of Process-Based Estimation 515
25.6.6 An Example of Estimation Using Use Case Points 517
25.6.7 Reconciling Estimates 518
25.6.8 Estimation for Agile Development 519
25.7 Project Scheduling 520
25.7.1 Basic Principles 521
25.7.2 The Relationship Between People and Effort 522
25.8 Defining a Project Task Set 523
25.8.1 A Task Set Example 524
25.8.2 Refinement of Major Tasks 524
25.9 Defining a Task Network 525
25.10 Scheduling 226
25.10.1 Time-Line Charts 526
25.10.2 Tracking the Schedule 528
25.11 Summary 530

CHAPTER 26 RISK MANAGEMENT 532

26.1 Reactive Versus Proactive Risk Strategies 533
26.2 Software Risks 534
26.3 Risk Identification 535
26.3.1 Assessing Overall Project Risk 536
26.3.2 Risk Components and Drivers 537
26.4 Risk Projection 538